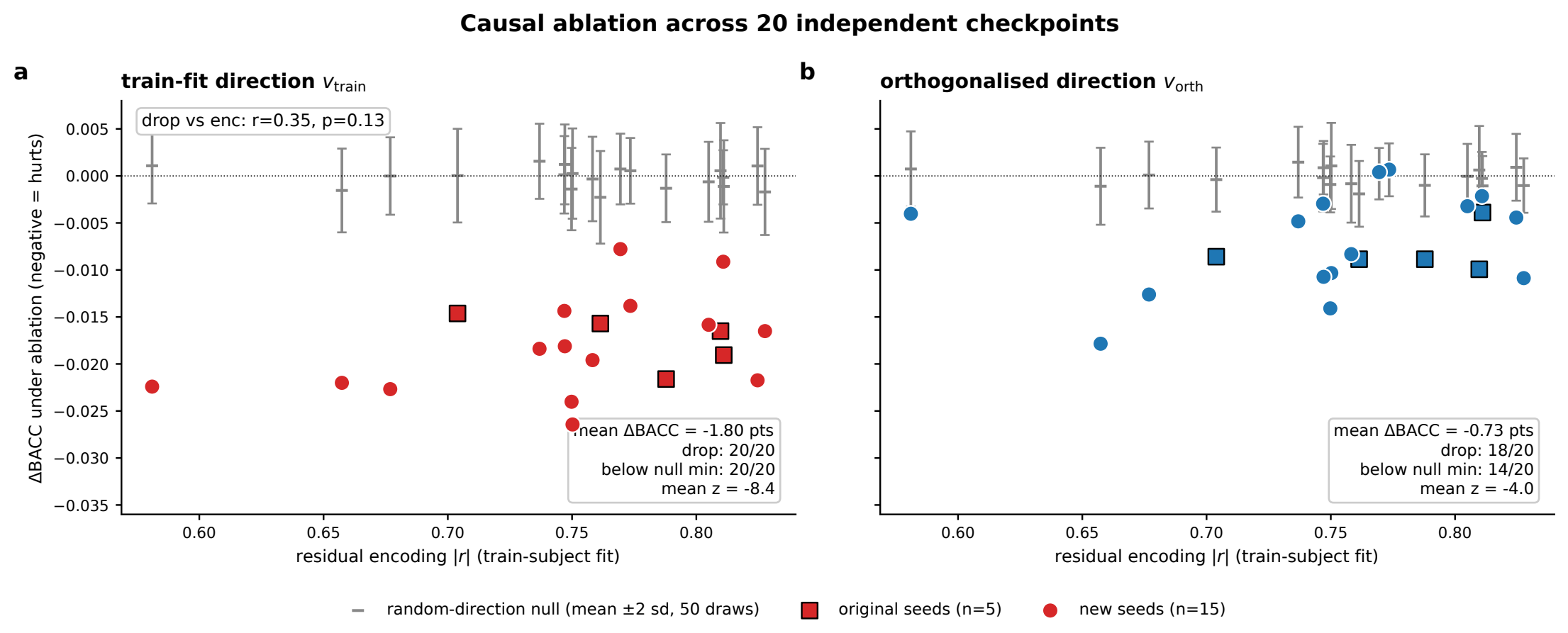
Causal ablation across 20 independent checkpoints
a
train-fit direction v_{train}
drop vs enc: r=0.35, p=0.13
mean ΔBACC = -1.80 pts
drop: 20/20
below null min: 20/20
mean z = -8.4
b
orthogonalised direction v_{orth}
mean ΔBACC = -0.73 pts
drop: 18/20
below null min: 14/20
mean z = -4.0
ΔBACC under ablation (negative = hurts)
0.005
0.000
−0.005
−0.010
−0.015
−0.020
−0.025
−0.030
−0.035
0.60
0.65
0.70
0.75
0.80
residual encoding |r| (train-subject fit)
random-direction null (mean ±2 sd, 50 draws)
original seeds (n=5)
new seeds (n=15)